

Sensor Threat Isolation for Cyber-Physical Systems[★]

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Abstract: This paper considers the isolability of sensor faults and malicious cyber attacks in cyber-physical systems. Attacks and faults are characterized by their basis functions, which are utilized for threat isolation and to characterize their isolability. An isolability metric is proposed that exploits the threat basis function to quantify the threat isolability. It is rigorously proved that a threat is non-isolable if and only if this metric is zero. A methodology to analytically calculate the proposed isolability metric is developed by solving a minimax optimization problem. Our findings are validated with numerical simulations.

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1. INTRODUCTION

The rising complexity of cyber-physical systems (CPS) makes them more vulnerable to malicious cyber attacks (Cardenas et al. (2008)). Apart from cyber attacks, traditional physical component faults significantly affect the CPS. Hence, the development of threat isolation techniques to identify the occurring threat type (cyber attacks or physical faults) becomes highly important. Fault isolation has been extensively studied in the fault diagnosis literature such as Ding (2008); Zhang et al. (2005). Fault isolation typically refers to *fault location* isolation, aiming to identify the faulty components or the type of fault. In addition, by considering integrity cyber attacks, *attack identification* aims to identify the attacked sensors and actuators, and also belongs to a type of location isolation Pasqualetti et al. (2013). However, typical *fault location* isolation and *attack identification* cannot effectively address the threat isolation problem since a cyber attack and a physical fault may occur at the same location from the system perspective (e.g., the same sensor channels).

The problem of threat isolation has not been adequately studied so far. Chantbery and Subias (2019) is one of the first works to consider the threat isolation problem, using

relevant simulation results. Our previous works in Zhang et al. (2021c,a, 2022) explore threat isolation issues by considering simple threat scenarios. In Zhang et al. (2021c), an isolation scheme for distinguishing physical faults from occurring threats is designed, while in Zhang et al. (2021a), only constant sensor faults are considered. In addition, some specific methodologies, such as the backward-in-time signal processors in Zhang et al. (2021c,a) and the watermarking technique in Zhang et al. (2021a), are developed for threat isolation purposes. However, the literature lacks a threat isolability framework that would help determine when a threat is isolable.

Contribution: This paper investigates the problem of threat isolation between physical sensor faults and cyber attacks. In particular, we establish a threat isolability framework to characterize the isolability of sensor faults, cyber attacks and their simultaneous presence. To achieve isolation, we make use the time-dependent basis functions that characterize the distinct time evolution features of faults and attacks. Also, the minimax approximation theory is used to define an isolability metric. Specifically, the *threat isolability* is characterized by using the threat basis functions. We derive a theoretical condition that determines when a threat (attack or fault) is non-isolable (cannot be isolated). Moreover, for each threat, an isolability *metric* is proposed to quantify its isolability, which has the property that a threat is more difficult to isolate if its isolability metric is more close to zero. It is rigorously proved that a threat is non-isolable if and only if the proposed isolability metric is zero. In addition, a methodology

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to analytically calculate the isolability metric is developed by solving a minimax optimization problem.

Notation: The notation $|\cdot|$ is used to represent the absolute value for scalars, and the 2-norm for vectors and matrices. The $\mathcal{C}(T, \mathbb{R})$ denotes the space of scalar-valued continuous functions from the compact interval T to the Euclidean space $\mathbb{R}$. A set of scalar-valued functions $\{f_1(t), \dots, f_n(t)\}$ with $f_i(t) \in \mathcal{C}(T, \mathbb{R})$ for $i \in \{1, \dots, n\}$ is *linearly independent* if $k_1 f_1(t) + \dots + k_n f_n(t) = 0$ for all $t \in T$ if and only if $k_1 = \dots = k_n = 0$. In addition, a set of scalar-valued functions $\{f_1(t), \dots, f_n(t)\}$ with $f_i(t) \in \mathcal{C}(T, \mathbb{R})$ for $i \in \{1, \dots, n\}$ spans a subspace $\mathcal{F}$, i.e., $\mathcal{F} = \text{span}\{f_1(t), \dots, f_n(t)\}$, if for any function $f(t) \in \mathcal{F}$, there exist scalars $k_1, \dots, k_n$ such that $f(t) = k_1 f_1(t) + \dots + k_n f_n(t)$. A set of scalar-valued functions $\{f_1(t), \dots, f_n(t)\}$ with $f_i(t) \in \mathcal{C}(T, \mathbb{R})$ for $i \in \{1, \dots, n\}$ is a basis of the subspace $\mathcal{F}$ if it is linearly independent and if it spans $\mathcal{F}$, and we say that each element in this basis is a basis function. In addition, $\mathcal{F}$ is a n -dimensional subspace of $\mathcal{C}(T, \mathbb{R})$ if its basis has n basis functions and each basis function belongs to $\mathcal{C}(T, \mathbb{R})$. For a function $f(t) = [f_1(t), \dots, f_n(t)] \in \mathbb{R}^{1 \times n}$, a basis of $f(t)$ refers to a basis of the subspace spanned by $\{f_1(t), \dots, f_n(t)\}$. We use the acronym “w.r.t.” to represent “with respect to”.

2. PROBLEM FORMULATION

We consider a typical CPS, depicted in Fig. 1, consisting of a physical plant, actuator and sensor data communication networks, a controller and an anomaly detector. The physical plant considered in this paper is a single output system. The closed-loop system in the nominal case (no attacks or faults) is given as follows:

$$\dot{x}_n = Ax_n + Bu_n, \quad u_n = Ky_n, \quad y_n = Cx_n, \quad (1)$$

where $x_n \in \mathbb{R}^{n_p}$, $Cx_n \in \mathbb{R}$ and $u_n \in \mathbb{R}^{n_u}$ are the state of the physical plant, the sensor measurement and the received control input by the actuator from the actuator communication network, respectively. The signal $y_n \in \mathbb{R}$ is the system output, received by the controller and detector via the sensor communication network. The control signal $Ky_n \in \mathbb{R}^{n_u}$ and the sensor measurement $Cx_n \in \mathbb{R}$ are transmitted by the actuator and sensor communication networks respectively. Moreover, $A \in \mathbb{R}^{n_p \times n_p}$, $B \in \mathbb{R}^{n_p \times n_u}$, $K \in \mathbb{R}^{n_u}$ and $C \in \mathbb{R}^{1 \times n_p}$ are known matrices/vectors by the defender, $A + BKC$ is a Hurwitz matrix, and (A, C) is observable.

2.1 Considered Threats

Cyber stealthy integrity attacks and physical faults are considered in this paper, which have distinct time evolution features. Basis function is a concept in approximation theory used for constructing approximations (see, e.g., Cheney (1998); Farrell and Polycarpou (2006)). Borrowing this concept, we show in the sequel that the time evolution properties of some cyber attacks and physical faults can be characterized by suitable basis functions. Below, we explain how attacks and faults can be *linearly parameterized* by using basis functions. To this end, we let x , u and y represent the state, received control input and sensor measurements in the anomaly case, respectively, corresponding to x_n , u_n and y_n in the nominal case.

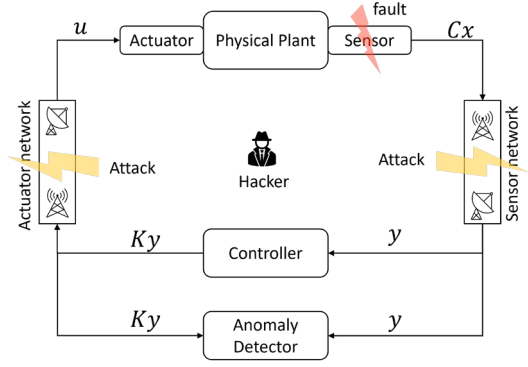


Fig. 1. Diagram of CPS in the presence of potential cyber attacks and physical sensor faults.

Stealthy integrity attacks Cyber stealthy integrity attacks can simultaneously disrupt CPS and maintain stealthiness by utilizing system model knowledge in an intelligent manner (see, e.g., Zhang et al. (2021a); Teixeira et al. (2015); Pasqualetti et al. (2013)). The latter motivates the research community to develop suitable approaches to detect and identify them. Fortunately, such attacks, that are highly reliant on the underlying system model, result in unique effects on system outputs that can be exploited by the defender.

It should be noted that attack effects on the system output are of high importance for threat detection and isolation. Instead of studying the attack signals, *we consider the attack effects on the system output*. Hence, the notation $a(t)$ used in the sequel represents the effects caused by the attack rather than the true attack signals. We use T_a to represent the time when the attack occurs.

Replay Attacks are equipped with recording and replaying functionalities (Mo and Sinopoli (2009)). We consider that the adversary first records Cx_n , communicated through the sensor communication network, starting at time $T_a - \kappa_a$ for a recording period κ_a . Then, the replaying procedure starts at T_a and ends at $T_a + \kappa_a$. Hence, the replay attack signal (i.e., the true attack signal) is $y_n(t - \kappa_a)$, and the attack model can be written as

$$\dot{x}'_n = Ax'_n + BKy'_n, \quad y'_n = Cx'_n, \quad (2)$$

where $x'_n(t) \triangleq x_n(t - \kappa_a)$ and $y'_n(t) \triangleq y_n(t - \kappa_a)$. Similar with Zhang et al. (2021c), by defining an attack effect a as the deviation between the outputs in the nominal and attack cases, i.e., $a(t) \triangleq y'_n(t) - Cx(t)$ for $t \in [T_a, T_a + \kappa_a]$, the system (1) under the replay attack can be described by

$$\dot{x} = Ax + BKy, \quad y = Cx + a(t), \quad (3)$$

where $y(t) = y'_n(t)$ for $t \in [T_a, T_a + \kappa_a]$. From (2) and (3), we have $\dot{x}'_n - \dot{x} = A(x'_n - x)$. Hence, $a(t)$ can be linearly parameterized as

$$a(t) = F^a(t)\theta^a, \quad \forall t \in [T_a, T_a + \kappa_a], \quad (4)$$

where $F^a(t) = Ce^{A(t-T_a)}$ is the known basis, and $\theta^a = x'_n(T_a) - x(T_a)$ is the unknown attack parameter vector.

Zero-dynamics and Covert Attacks are referred to as *undetectable attacks* in Pasqualetti et al. (2013) and stealthy integrity attacks in Zhang et al. (2021b). In an attack event, the false data a_u and a_y are cooperatively

injected into the actuator and sensor communication networks respectively such that the effects on the system output are minimized. Under such an attack, system (1) is written as follows:

$$\dot{x} = Ax + BKy + B_a a_u, \quad y = Cx + D_a a_y, \quad (5)$$

where B_a and D_a are the attack distribution matrices with respect to actuator and sensor communication networks. Based on Pasqualetti et al. (2013), the attack pair $(B_a a_u, D_a a_y)$ can be intelligently designed such that

$$y(t, T_a, x_n(T_a), a_u, a_y) = y_n(t, T_a, x_n(T_a) + \Delta x_0, 0, 0), \quad (6)$$

for some $\Delta x_0 \in \mathbb{R}^{n_p}$, where the left-hand side represents the output of system (5) with the initial condition $x_n(T_a)$, and the right-hand side represents the output of system (1) with the initial condition $x_n(T_a) + \Delta x_0$. Note that for the linear system (1), $y_n(t, T_a, x_n(T_a) + \Delta x_0, 0, 0) = y_n(t, T_a, x_n(T_a), 0, 0) + y_n(t, T_a, \Delta x_0, 0, 0)$ where the second term can be explicitly written as $Ce^{(A+BKC)(t-T_0)} \Delta x_0$. By defining a as the change in output due the attack, i.e., $a(t) \triangleq y(t, T_a, x_n(T_a), a_u, a_y) - y_n(t, T_a, x_n(T_a), 0, 0) = y_n(t, T_a, \Delta x_0, 0, 0)$, the closed-loop system that is able to generate $y(t, T_a, x_n(T_a), a_u, a_y)$ can be written as in (3) where

$$a(t) = F^a(t)\theta^a, \quad \forall t > T_a, \quad (7)$$

with the continuous basis $F^a = Ce^{(A+BKC)(t-T_a)}$ known by the defender and the unknown parameter $\theta^a = \Delta x_0$.

Therefore, the effects $a(t)$ of replay attacks and zero-dynamics and covert attacks can be linearly parameterized with $F^a(t)$ depending only on the system knowledge A, B, K known by the defender. Apart from the aforementioned attacks, various other types of stealthy integrity attacks, such as false data injection attacks and bias injection attacks in Teixeira et al. (2015), can also be expressed in the form of (4) or (7) with known basis $F^a(t)$. Note that different attacks may have different bases. Hence, in this paper, we consider the attacks that their effects on the system output are linearly parameterized as follows:

$$a(t) = F^a(t)\theta^a, \quad \forall t \geq T_a, \quad (8)$$

where $\theta^a \in \mathbb{R}^{n_a}$ is the unknown parameter and $F^a(t) \in \mathbb{R}^{1 \times n_a}$ is the continuous basis known by the defender.

Physical faults In terms of faults, we consider sensor faults that appear in the output channel, making their isolation from attacks more challenging. Faults can be linearly parameterized with known basis functions derived based on the physical properties of faulty components (see, e.g., Burkholder and Tao (2011); Deb et al. (2022)). Also, faults can be well-approximated by a set of basis functions such as polynomial, sigmoidal and Gaussian functions based on the data obtained from experiment tests.

In this paper, we consider that the fault is unknown and approximated over a finite time interval by a set of preselected basis functions with unknown parameters such as a polynomial, a neural network or a Fourier series (for periodic function). We use f and T_f to denote the fault and its time of occurrence, and $[T_f, T_f + \kappa_f)$ as the concerned finite time interval where $\kappa_f > 0$. Then, sensor faults considered in this paper can be linearly parameterized as

$$f(t) = F^f(t)\theta^f, \quad \forall t \in [T_f, T_f + \kappa_f), \quad (9)$$

where $\theta^f \in \mathbb{R}^{n_f}$ is the unknown parameter and $F^f(t) \in \mathbb{R}^{1 \times n_f}$ the known continuous basis by the defender.

Given cyber attacks and physical faults, three potential threat scenarios may develop in the system: (i) *attacks occur at time T_a but no sensor faults occur*, (ii) *sensor faults occur at time T_f but no attacks occur*, and (iii) *attacks and sensor faults occur simultaneously at time T_{fa}* . Those cases will be referred to as *threat a* , *threat f* and *threat $a \& f$* , respectively. We let $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}$ and T_0 represent the threat function with respect to time and the threat occurring time respectively, satisfying

$$(T_0, \phi) \in \{(T_a, a), (T_f, f), (T_{af}, a + f)\}. \quad (10)$$

In the presence of the threats, the system is described by

$$\dot{x} = Ax + Bu, \quad u = Ky, \quad (11a)$$

$$y = Cx + \beta(t - T_0)\phi(t), \quad (11b)$$

where $\beta : \mathbb{R} \rightarrow \{1, 0\}$ is a step function, i.e., $\beta(t - T_0) = 1$ for $t \geq T_0$ and 0, otherwise, and describes the time profile of the occurring threat by indicating its occurrence at T_0 .

2.2 Objectives

The bases $F^a(t)$ and $F^f(t)$ characterize the time evolution features of the attack and fault respectively. The objectives of this paper are presented below:

- (i) Provide a theoretical characterization of threat isolability by using the bases $F^a(t)$ and $F^f(t)$;
- (ii) Develop a metric to quantitatively characterize threat isolability based on the bases $F^a(t)$ and $F^f(t)$, and propose an analytical approach for its calculation.

Objective (i) aims to provide the theoretical tools to deduce when threats are isolable, by utilizing their corresponding basis functions. In addition, objective (ii) aims to provide an algorithmic approach to determine whether a pair of threats is isolable.

3. THREAT ISOLATION

In this section, we provide the theoretical characterization of threat isolability using $F^a(t)$ in (8) and $F^f(t)$ in (9). To this end, we let T_d be the time when an anomaly (threat) is detected by an equipped detector with $T_d > T_0$. In addition, we suppose that the state value of system (11) at T_d is known by the defender, and denoted as x_0 .

Definition 1. (non-isolable Threat). Consider system (11) with the potential threat ϕ satisfying (10), and the attacks and faults satisfying (8) and (9) respectively. We call the threat $i \in \{a, f\}$ non-isolable if for any $\theta^i \in \mathbb{R}^{n_i}$ there exists a matrix $F^{j,i} \in \mathbb{R}^{n_j \times n_i}$ such that $y(x_0, F^i(t)\theta^i, t) = y(x_0, F^j(t)F^{j,i}\theta^i, t)$, $j = \{a, f\} \setminus \{i\}$ for all $t \geq T_d$. ■

Intuitively, the threat i is non-isolable if there exists a threat j that yields the same output with threat i . Based on the non-isolability condition in Definition 1, the threat $a \& f$ is non-isolable if and only if there exists an $i \in \{a, f\}$ such that the threat i is non-isolable. Before presenting the main result, we introduce the following lemma and the definition of a (left) invertible system.

Lemma 1. Zhang et al. (2021a) For the threat $i \in \{a, f\}$, consider the auxiliary system given as follows:

$$\dot{\Omega}^i = A_0 \Omega^i + BKF^i, \quad \Omega^i(T_d) = 0, \quad \forall i \in \{a, f\}, \quad (12)$$

where $\Omega^i \in \mathbb{R}^{n_p \times n_i}$ and $A_0 \triangleq A + BKC$. Then, in the attack a case and fault f case, the state x of system (11) satisfies $x - \Omega^a \theta^a = x - \Omega^f \theta^f$. Moreover, by letting

$$y_{\Omega}^i \triangleq C\Omega^i + F^i, \bar{x} \triangleq x - \Omega^i\theta^i, \quad (13)$$

when threat i occurs, y in (11) can be written as

$$\dot{\bar{x}} = A_0\bar{x}, \bar{x}(T_d) = x_0, \quad (14a)$$

$$y = C\bar{x} + y_{\Omega}^i\theta^i, \forall i \in \{a, f\}. \quad (14b)$$

We formulate (12) and y_{Ω}^i in (13) as a new system with matrices A_0, BK, C and 1, where Ω^i is the state, F^i is the input and y_{Ω}^i is the output, which is referred to as system $(A_0, BK, C, 1)$. The definition of (left) invertible in the context of system $(A_0, BK, C, 1)$ is given as follows.

Definition 2. (Invertible System (Moylan, 1977)). Let F^s and F^h ($s \neq h$) be any two inputs of system $(A_0, BK, C, 1)$, and let y_{Ω}^s and y_{Ω}^h be the corresponding outputs (for $\Omega^s(T_d) = \Omega^h(T_d) = 0$), system $(A_0, BK, C, 1)$ is (left) invertible if $y_{\Omega}^s(t) = y_{\Omega}^h(t)$ for all $t \geq T_d$ implies that $F^s(t) = F^h(t)$ for all $t \geq T_d$. ■

A large variety of systems are left invertible such as vehicle chassis systems in Wang et al. (2018). Then, our main threat isolability result is presented below.

Theorem 1. (Threat Isolability w.r.t. Basis). Consider system (11) and suppose that system $(A_0, BK, C, 1)$ is (left) invertible. Then, the following statements are equivalent:

- (i) the threat $i \in \{a, f\}$ is non-isolable for $t \geq T_d$;
- (ii) there exists a matrix $F^{j,i} \in \mathbb{R}^{n_j \times n_i}$ such that $F^i(t) = F^j(t)F^{j,i}$, $t \geq T_d$ where $F^i(t)$ and $F^j(t)$ are the bases of the threat i and j respectively for $i, j \in \{a, f\}$. ■

Theorem 1 characterizes the threat isolability through a condition on non-isolable threats. According to Theorem 1, the only factor for the isolability of the threat i is the existence of $F^{j,i}$ satisfying result (ii) in Theorem 1. In the next section, we develop a methodology to analytically calculate the “best” $F^{j,i}$ by solving a suitable optimization problem that minimizes $F^i(t) - F^j(t)F^{j,i}$.

4. THREAT ISOLABILITY METRIC

In this section, we propose a threat isolability metric, develop its calculation methodology and present the threat isolation decision process. It should be noted that we consider a finite time interval $T \triangleq [T_d, T_b] \subset [T_d, \infty)$ with $T_b > T_d$ instead of considering an infinite time horizon.

4.1 Metric of Threat Isolability

In this subsection, the threat isolability metric in the finite time interval T is proposed and its characteristic to quantify threat isolability is rigorously analyzed. We start by proposing the metric that quantifies the isolability of the threat i on T , denoted by η^i , as follows:

$$\eta^i \triangleq \sum_{s=1}^{n_i} \text{dist}(F_s^i(t), F^j(t)), j = \{a, f\} \setminus \{i\}, \quad (15)$$

where $t \in T$, $\eta^i \in \mathbb{R}$ is a constant and F_s^i refers to the s -th column of F^i . The $\text{dist}(F_s^i(t), F^j(t))$ represents a distance from F_s^i to the space spanned by the basis F^j on T , i.e., $\text{span}\{F_1^j, \dots, F_{n_j}^j\}$, and is defined as follows:

$$\text{dist}(F_s^i(t), F^j(t)) \triangleq \min_{c_s \in \mathbb{R}^{n_j}} \max_{t \in T} |F_s^i(t) - F^j(t)c_s|, t \in T, \\ j = \{a, f\} \setminus \{i\}, \forall i \in \{a, f\}, \forall s \in \{1, \dots, n_i\}, \quad (16)$$

where $c_s = [c_{s,1}, \dots, c_{s,n_j}]^T \in \mathbb{R}^{n_j}$ is the coefficient vector. Intuitively, a small η^i means that $F^i(t)$ has a slight difference with $F^j(t)$, which implies that the threat i is very similar with the threat j , and thus is more difficult to isolate. Especially, $\eta^i = 0$ when the threat i is completely non-isolable. In addition, it should be noted that other types of norms such as 2-norm can also be used to characterize the distance $\text{dist}(F_s^i(t), F^j(t))$. The following theorem characterizes the property of metric η^i .

Theorem 2. (Threat Isolability w.r.t. Metric η^i). Consider system (11) and suppose that $(A_0, BK, C, 1)$ is (left) invertible. Then, the following statements are equivalent:

- (i) the threat $i \in \{a, f\}$ is non-isolable;
- (ii) the isolability metric η^i of the threat i equals to zero;
- (iii) for any $s \in \{1, \dots, n_i\}$, the distance from $F_s^i(t)$ to $\text{span}\{F_1^j(t), \dots, F_{n_j}^j(t)\}$ on T equals to zero, i.e.,

$$\text{dist}(F_s^i(t), F^j(t)) = 0, \forall s \in \{1, \dots, n_i\}, \quad (17)$$

where $F^i(t)$ and $F^j(t)$ are the bases of the threat i and j respectively. ■

According to Theorem 2, η^i in (15) can quantify the threat isolability based on the following *decision principle*: the threat $i \in \{a, f\}$ is non-isolable if and only if $\eta^i = 0$ (or equivalently, is isolable if and only if $\eta^i > 0$), and the threat a & f is isolable if and only if $\eta^a > 0$ and $\eta^f > 0$.

4.2 Threat Isolability Metric Calculation

The $\text{dist}(F_s^i(t), F^j(t))$ defined in (16) is the absolute value of the approximation error of the best approximation (defined in Powell (1981)) from the space $\text{span}\{F_1^j, \dots, F_{n_j}^j\}$ to F_s^i over the time interval T . Furthermore, (16) defines a minimax optimization problem. In this subsection, we present our approach to analytically obtain the best approximation by solving the minimax optimization problem (16). Note that the results are presented based on the notation in (15) and (16), and the indexes $i, j \in \{a, f\}$, $i \neq j$. Let $\mathcal{F}^j$ denote the subspace spanned by the basis $F^j(t)$, i.e.,

$$\mathcal{F}^j \triangleq \text{span}\{F_1^j(t), \dots, F_{n_j}^j(t)\}.$$

Then, for any $F_s^i(t) \in \mathcal{C}(T, \mathbb{R})$ with $i \in \{a, f\}$ and $s \in \{1, \dots, n_i\}$, the approximation from the space $\mathcal{F}^j$ to $F_s^i(t)$ can be described by

$$\rho_s(t, c_s) \triangleq \sum_{k=1}^{n_j} c_{s,k} F_k^j(t), \forall s \in \{1, \dots, n_i\},$$

where $c_s = [c_{s,1}, \dots, c_{s,n_j}]^T \in \mathbb{R}^{n_j}$ is the coefficient vector. Also, the approximation error is given as follows:

$$\gamma_s(t, c_s) \triangleq F_s^i(t) - \rho_s(t, c_s), \forall s \in \{1, \dots, n_i\}.$$

Then, we have $\gamma_s(t, c_s) \in \mathcal{C}(T, \mathbb{R})$ for any $c_s \in \mathbb{R}^{n_j}$.

An approximation $\rho_s(t, c_s^*)$ with a specific coefficient vector c_s^* is referred to as the *best approximation* from $\mathcal{F}^j$ to $F_s^i(t)$, and $c_s^* \in \mathbb{R}^{n_j}$ is the *best coefficient vector* if the approximation $\rho_s(t, c_s^*)$ minimizes $|\gamma_s(t, c_s)|$ over $c_s \in \mathbb{R}^{n_j}$ and $t \in T$. Hence, $\text{dist}(F_s^i(t), F^j(t))$ in (16) can be represented by the following minimax problem:

$$\text{dist}(F_s^i(t), F^j(t)) = z^* \triangleq \min_{c_s \in \mathbb{R}^{n_j}} \max_{t \in T} |\gamma_s(t, c_s)|, \quad (18a)$$

$$c_s^* \triangleq \arg \min_{c_s \in \mathbb{R}^{n_j}} \max_{t \in T} |\gamma_s(t, c_s)|, \quad (18b)$$

where z^* is the optimal value. Before presenting the solution to (18), we make the following assumption.

Assumption 1. For any $j \in \{a, f\}$, the basis $F^j(t)$ satisfies the *Haar condition* on T , i.e., the basis functions $F_1^j(t), \dots, F_{n_j}^j(t)$ are continuous, and for any set of $\{t_1, \dots, t_{n_j}\}$ satisfying $T_d \leq t_1 < t_2 < \dots < t_{n_j} \leq T_b$, the row vectors $[F_1^j(t_s), \dots, F_{n_j}^j(t_s)]$ for $s \in \{1, \dots, n_j\}$ are linearly independent. $\blacktriangle$

The *Haar condition* guarantees the uniqueness of the best approximation Cheney (1998), and is widely used in the approximation literature Cheney (1998); Powell (1981). Many types of bases such as $\{1, t, t^2\}$ on $[0, 1]$ and $\{1, e^{-t}, e^{-2t}\}$ on $[0, 1]$ satisfy the *Haar condition*. A theorem to provide the analytical values of z^* and c_s^* is presented below.

Theorem 3. Consider the subspace $\mathcal{F}^j$ with its basis satisfying Assumption 1, and the function $F_s^i(t) \in \mathcal{C}(T, \mathbb{R})$ with $s \in \{1, \dots, n_i\}$. In addition, consider $t_1^*, \dots, t_{n_j+1}^*$ satisfying $T_d \leq t_1^* < \dots < t_{n_j+1}^* \leq T_b$ and $\gamma_s(t_h^*, c_s^*) = -\gamma_s(t_{h+1}^*, c_s^*) = \sigma_h^* z^*$ for any $h \in \{1, \dots, n_j\}$ where $\sigma_h^* = \text{sign}(\gamma_s(t_h^*, c_s^*))$. Then, the optimal value z^* in (18a) and the best coefficient vector c_s^* in (18b) satisfy

$$\begin{bmatrix} z^* \\ c_s^* \end{bmatrix}^T = G^{-1}b, \quad (19)$$

where

$$G = \begin{bmatrix} 1 & \sigma_1^* F_1^j(t_1^*) & \dots & \sigma_1^* F_{n_j}^j(t_1^*) \\ 1 & \sigma_2^* F_1^j(t_2^*) & \dots & \sigma_2^* F_{n_j}^j(t_2^*) \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \sigma_{n_j+1}^* F_1^j(t_{n_j+1}^*) & \dots & \sigma_{n_j+1}^* F_{n_j}^j(t_{n_j+1}^*) \end{bmatrix},$$

$$b = [\sigma_1^* F_s^i(t_1^*) \quad \sigma_2^* F_s^i(t_2^*) \quad \dots \quad \sigma_{n_j+1}^* F_s^i(t_{n_j+1}^*)]^T.$$

Remark 1. Under Assumption 1, G in (19) is invertible. 3 enables the calculation of $\text{dist}(F_s^i(t), F^j(t))$ in (16) for any $s \in \{1, \dots, n_i\}$ and for any $i \in \{a, f\}$, thereby enabling the calculation of the isolability metric η^i in (15). Hence, the isolability of the threats a , f and $a \& f$, can be judged analytically based on the isolation decision principle presented in the previous subsection. ∇

The optimal c_s^* and z^* are specified in Theorem 3 under a given set of $\bar{t}^* \triangleq \{t_1^*, \dots, t_{n_j+1}^*\}$. Below, we provide conditions to verify the optimality of $\bar{t}^*$. To this end, we define a convex region Π such that $\bar{t}^* \in \Pi$, as follows:

$$\Pi \triangleq [T_d, T_b] \times (T_d, T_b) \times \dots \times (T_d, T_b) \times (T_d, T_b]. \quad (20)$$

The optimality conditions for $\bar{t}^*$ are presented below.

Theorem 4. (First-order Sufficient Condition). Let the basis functions $F_1^j(t), \dots, F_{n_j}^j(t)$ and $F_s^i(t)$ be continuously differentiable with respect to time t . Then, $\bar{t}^*$ is the set of the optimal times of the minimax problem (18) if for each feasible direction $d \neq 0$ in Π at $\bar{t}^*$ and for each $h \in \{1, \dots, n_j + 1\}$,

$$d^T \nabla_{\bar{t}} \sigma_h^* \gamma_s(t_h^*, c_s^*(\bar{t}^*)) \leq -\delta |d|, \quad (21)$$

where $\delta > 0$ is any scalar, and

$$\nabla_{\bar{t}} \sigma_h^* \gamma_s(t_h, c_s^*(\bar{t})) \triangleq \left[\frac{\partial \sigma_h^* \gamma_s(t_h, c_s^*(\bar{t}))}{\partial t_1}, \dots, \frac{\partial \sigma_h^* \gamma_s(t_h, c_s^*(\bar{t}))}{\partial t_{n_j+1}} \right]^T. \quad (22)$$

Remark 2. Theorems 3 and 4 provide conditions that enable a set of optimal values for c_s^*, z^* for given $\bar{t}^*$ and verify the optimality of $\bar{t}^*$ respectively. Therefore, they provide a means to validate whether the distance between two sets of basis functions is optimal. These results can be used in the design of specific isolators, ensuring that the obtained values for $\eta^i, i \in \{a, f\}$ are optimal. In addition, Theorem 4 provides sufficient conditions to verify the optimality of $\bar{t}^*$. This set of values may be obtained by applying suitable numerical calculation algorithms, such as the *Remes Algorithm* in Cheney (1998). ∇

5. NUMERICAL EXAMPLE

In this section, an illustrative example is presented. We consider the following matrices for system (1):

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}, B = \begin{bmatrix} -1.67 & 0 & 0 \\ 0 & -1.93 & 0 \\ 0 & 0 & 1 \end{bmatrix}, K = \begin{bmatrix} -1.1946 & 0.6858 & 0.2919 \end{bmatrix},$$

and $C = [1, 0, 0]$. It can be verified that system $(A_0, BK, C, 1)$ is left invertible and satisfies the requirement in Theorems 1 and 2. In addition, we consider that a threat is detected by some anomaly detector at $T_d = 50.2s$.

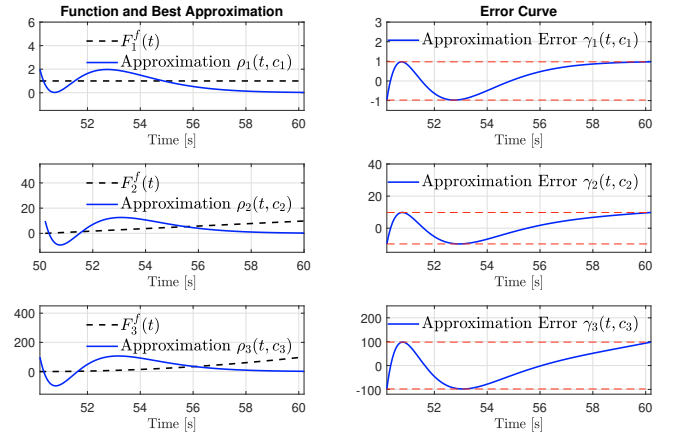


Fig. 2. Best approximation of $F^f(t)$ using the basis $F^a(t)$.

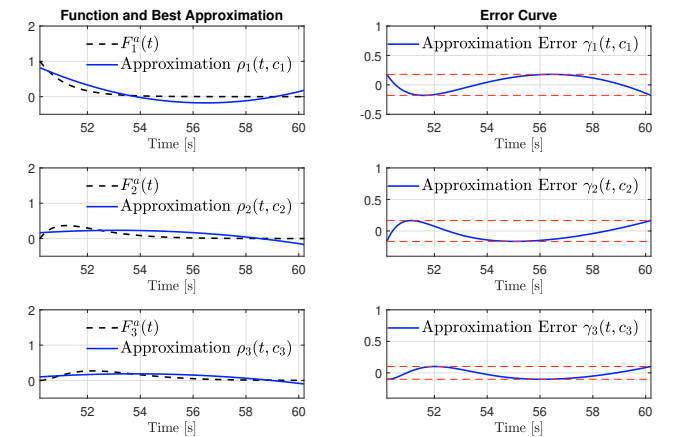


Fig. 3. Best approximation of $F^a(t)$ using the basis $F^f(t)$.

In this example, a replay attack and a sensor bias fault are considered. Regarding the replay attack, the attacker records the data at $T_a - \kappa_a = 20s$ and replays the data at $T_a = 50s$. The basis of this replay attack is a basis of $C \exp(A(t - T_a))$ and is given by $F^a(t) = [\exp(T_a - t), (t -$

$T_a) \exp(T_a - t), (\exp(T_a - t)(T_a - t)^2)/2]$. The fault occurs at $T_f = 20$ s and is represented by the basis $F^f(t) = [1, (t - T_f), (t - T_f)^2]$. Thus, $F^a(t)$ and $F^f(t)$ are continuous and satisfy the *Haar condition* in Assumption 1.

Now, we calculate the isolability metrics η^a and η^f based on Theorems 3 and 4. The time interval T is $[50.2, 60.2]$. By using the numerical *Remes Algorithm* in Cheney (1998), the value of $\bar{t}^*$ is first searched numerically over T , and then c_s^* and z^* are calculated based on Theorem 3. Then, $\bar{t}^*$ is verified based on Theorem 4 using three particular directions: $d_1 = [1, 0, 0, 0]^T$, $d_2 = [0, 0, 0, -1]^T$ and $d_3 = [1, 0, 0, 1]^T$. One calculation example that F^f approximates F_1^a is given as follows: by using *Remes Algorithm*, we obtain $\bar{t}^* = [50.2, 51.5956, 56.4245, 60.2]$, and then by using Theorem 3, we obtain $c_1^* = [0.8215, -0.3187, 0.0254]^T$ and $z^* = 0.1785$. In addition, for all $h \in \{1, 2, 3\}$, $d_1^T \nabla_{\bar{t}} \sigma_h^* \gamma_1(t_h^*, c_1^*) = -0.193$, $d_2^T \nabla_{\bar{t}} \sigma_h^* \gamma_1(t_h^*, c_1^*) = -0.0144$ and $d_3^T \nabla_{\bar{t}} \sigma_h^* \gamma_1(t_h^*, c_1^*) = -0.2074$. Hence, the sufficient condition in Theorem 4 is guaranteed.

The best approximations and the error curves are shown in Fig. 2 and 3. Moreover, we obtain $\eta^a = 0.3439 > 0$ and $\eta^f = 36.1460 > 0$. Based on the calculated isolability metrics $\eta^a > 0$ and $\eta^f > 0$, and the *isolation decision principle*, we conclude that all the threat types are isolable.

In addition, to verify that the threat scenarios can be isolated by some threat isolation scheme, we simulate the threat isolation scheme in Zhang et al. (2021a). The result is shown in Fig. 4. The subfigures in the first row show that the occurring threat is an attack, and the fault is identified based on the second row. In addition, the third row identifies the fault and attack threat scenario. Hence, the three threat scenarios are isolable.

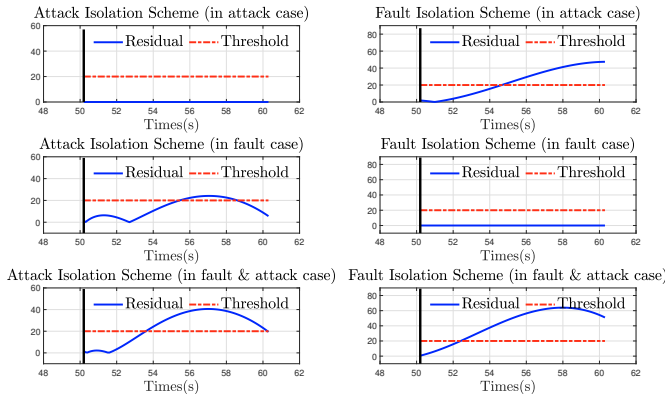


Fig. 4. Threat isolation for the three threat cases.

6. CONCLUSIONS

This paper has studied the isolability of three potential threat scenarios in cyber-physical systems: only cyber attacks, only sensor faults, and attack and fault co-existence. Threat isolability was characterized by using threat basis functions. Moreover, a metric was proposed to quantify the threat isolability, and a methodology to analytically calculate the isolability metric was developed. Finally, a numerical simulation example was given to validate the obtained results in the paper.

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